

Maths - Question 17

Original paper page 15; original question and option regions

If the coefficients of r^{th} , $(r+1)^{th}$ and $(r+2)^{th}$ terms in the expansion of $(1+x)^n$ are in the ratio of 4:15:42, then $n-r=$

$(1+x)^n$ యొక్క విస్తరణ యందు r వ, $(r+1)$ వ మరియు $(r+2)$ వ పదాల గుణకాలు 4:15:42 నిష్పత్తిలో ఉంటే $n-r=$

Options :

1. ✖

18

2. ✖

15

3. ✔

14

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Maths - Question 17 (continued)

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Maths - Question 18

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If the coefficients of $(2r + 6)^{th}$ and $(r - 1)^{th}$ terms in the expansion of $(1 + x)^{21}$ are equal, then the value of $r =$

$(1 + x)^{21}$ యొక్క విస్తరణలో $(2r + 6)$ మరియు $(r - 1)$ పదాల గుణకాలు సమానమైతే, r యొక్క విలువ =

Options :

1. ✖ 7

2. ✖ 5

3. ✔ 6

4. ✖ 8

Maths - Question 19

Original paper page 17; original question and option regions



If $\frac{13x+43}{2x^2+17x+30} = \frac{A}{2x+5} + \frac{B}{x+6}$ then $A + B =$

$\frac{13x+43}{2x^2+17x+30} = \frac{A}{2x+5} + \frac{B}{x+6}$ అయితే $A + B =$

Options :

1. ✓ 8

2. ✗ 18

3. ✗ 3

4. ✗ 5

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Maths - Question 20

Original paper page 17; original question and option regions

$$\tan \alpha + 2 \tan 2\alpha + 4 \tan 4\alpha + 8 \cot 8\alpha =$$

Options :

1. ✖

$$\sin \alpha$$

2. ✖

$$\cos \alpha$$

3. ✖

$$\tan \alpha$$

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Maths - Question 20 (continued)

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$\cot \alpha$

Maths - Question 21

Original paper page 18; original question and option regions

$$\tan 9^\circ - \tan 27^\circ - \tan 63^\circ + \tan 81^\circ =$$

Options :

1. ✓ 4

2. ✗ 3

3. ✗ 2

4. ✗ 1

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Maths - Question 22

Original paper page 18; original question and option regions

$$\cos 6^\circ \sin 24^\circ \cos 72^\circ =$$

Options :

$$-\frac{1}{8}$$

1. ✖

Maths - Question 22 (continued)

Original paper page 19; original question and option regions



$$-\frac{1}{4}$$

$$\frac{1}{8}$$

$$\frac{1}{4}$$

Maths - Question 23

Original paper page 19; original question and option regions

The values of x in $(-\pi, \pi)$ which satisfy the equation $8^{1+\cos^2 x + \cos^4 x + \dots} = 4^3$ are

$(-\pi, \pi)$ అంతరంలో ఉంటూ, $8^{1+\cos^2 x + \cos^4 x + \dots} = 4^3$ సమీకరణాన్ని తృప్తిపరచే x యొక్క విలువ

Options :

1. ✓

$$\pm \frac{\pi}{4}, \quad \pm \frac{3\pi}{4}$$

2. ✗

$$\pm \frac{\pi}{6}, \quad \frac{\pi}{3}$$

3. ✗

$$\pm \frac{\pi}{8}$$

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Maths - Question 23 (continued)

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$$\frac{\pi}{3}$$

Maths - Question 24

Original paper page 20; original question and option regions

$$\cot \left(\sum_{n=1}^{50} \tan^{-1} \left(\frac{1}{1+n+n^2} \right) \right) =$$

Options :

1. ✓ $\frac{26}{25}$

2. ✗ $\frac{25}{26}$

3. ✗ $\frac{50}{51}$

4. ✗ $\frac{52}{51}$

Maths - Question 25

Original paper page 21; original question and option regions



If $\sinh x = \frac{\sqrt{21}}{2}$ then $\cosh 2x + \sinh 2x =$

$\sinh x = \frac{\sqrt{21}}{2}$ అయితే $\cosh 2x + \sinh 2x =$

Options :

1. ✖ $\frac{21}{2}$

2. ✖ $\frac{25}{2}$

3. ✔ $\frac{23 + 5\sqrt{21}}{2}$

4. ✖ $\frac{32 + 5\sqrt{23}}{2}$

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Maths - Question 26

Original paper page 21; original question and option regions

In a ΔABC if $a = 13$, $b = 14$, $c = 15$ then $r_1 =$

ΔABC లో $a = 13$, $b = 14$, $c = 15$ అయితే $r_1 =$

Options :

1. ✖ $\frac{23}{2}$

Maths - Question 26 (continued)

Original paper page 22; original question and option regions



$$\frac{21}{2}$$

$$\frac{25}{2}$$

$$\frac{26}{3}$$

Maths - Question 27

Original paper page 22; original question and option regions

In a ΔABC , if $r:R:r_2 = 1:3:7$ then $\sin(A+C) + \sin B =$

ΔABC ແລະ $r:R:r_2 = 1:3:7$ ອັດຕະໂນ ຈົ່ງ ຄິດ $\sin(A+C) + \sin B =$

Options :

1. ✖ 0

2. ✖ $\sqrt{3}$

3. ✖ 1

4. ✔ 2

Question Number : 28 Question Id : 4509387548 Display Question Number : Yes Is Question

Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Inst

Time : 0

$$\text{In } \triangle ABC, (r_1 + r_2) \operatorname{cosec}^2 \frac{C}{2} =$$

$$\triangle ABC \text{ నందు } (r_1 + r_2) \operatorname{cosec}^2 \frac{C}{2} =$$

Options :

$$2R \cot^2 \frac{C}{2}$$

1. ✖

$$4R \tan^2 \frac{C}{2}$$

2. ✖

$$4R \cot^2 \frac{C}{2}$$

3. ✔

$$2R \tan^2 \frac{C}{2}$$

4. ✖

Maths - Question 30

Original paper page 24; original question and option regions

If the vectors $a\bar{i} + \bar{j} + 3\bar{k}$, $4\bar{i} + 5\bar{j} + \bar{k}$ and $4\bar{i} + 2\bar{j} + 6\bar{k}$ are coplanar, then $a =$

$a\bar{i} + \bar{j} + 3\bar{k}$, $4\bar{i} + 5\bar{j} + \bar{k}$ మరియు $4\bar{i} + 2\bar{j} + 6\bar{k}$ అను సదిశలు సతలీయాలైతే, $a =$

Options :

1. ✓

2

2. ✗

1

Maths - Question 30 (continued)

Original paper page 25; original question and option regions

3

4



Maths - Question 31

Original paper page 25; original question and option regions

Let $|\vec{a}| = 2, |\vec{b}| = 3$ and the angle between $\vec{a}$ and $\vec{b}$ be $\frac{\pi}{3}$. If a parallelogram is constructed with adjacent sides $2\vec{a} + 3\vec{b}$ and $\vec{a} - \vec{b}$, then its shorter diagonal is of length

$|\vec{a}| = 2, |\vec{b}| = 3$ మరియు $\vec{a}, \vec{b}$ ల మధ్య కోణము $\frac{\pi}{3}$ అనుకుందాం. $2\vec{a} + 3\vec{b}$ మరియు $\vec{a} - \vec{b}$ లు

ఆసన్న భుజాలుగా గల సమాంతర చతుర్భుజాన్ని నిర్మిస్తే చిన్న వికర్ణము యొక్క పొడవు

Options :

108

1. ✖

172

2. ✖

$6\sqrt{3}$

3. ✔

$2\sqrt{43}$

4. ✖

Mandatory : No Calculator : None Response Time : N.A Think Time : N.A 
Time : 0

The values of x for which the angle between the vectors $x^2\bar{i} + 2x\bar{j} + \bar{k}$ and $\bar{i} - 2\bar{j} + x\bar{k}$ is obtuse lie in the interval

$x^2\bar{i} + 2x\bar{j} + \bar{k}$ మరియు $\bar{i} - 2\bar{j} + x\bar{k}$ అను సదిశల మధ్య కోణము గురుకోణమయ్యేటట్లుగా ఉన్న x విలువలు ఉండే అంతరం

Options :

1. ✖ $(-\infty, 0) \cup (3, \infty)$

2. ✔ $(0, 3)$

3. ✖ $[0, 3]$

4. ✖ $(-\infty, 0) \cup [3, \infty)$

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Maths - Question 34

Original paper page 27; original question and option regions

Based on the following statements, choose the correct option

Statement-I: The variance of the first n even natural numbers is $\frac{n^2-1}{4}$

Statement-II: The difference between the variance of the first 20 even natural numbers and their arithmetic mean is 112

క్రింది ప్రవచనాల ఆధారంగా, సరియైన ఐచ్ఛికాన్ని ఎన్నుకోండి

ప్రవచనం-I: మొదటి n సరి సహజసంఖ్యల యొక్క విస్తృతి $\frac{n^2-1}{4}$

ప్రవచనం-II: మొదటి 20 సరి సహజసంఖ్యల విస్తృతి మరియు వాటి అంక మధ్యమముల మధ్య గల భేదం 112



Both Statements are true and II is a correct explanation of I

రెండు ప్రవచనాలు సత్యము మరియు I యొక్క సరియైన వివరణ II

Both Statements are true but II is not a correct explanation of I

రెండు ప్రవచనాలు సత్యము, కాని I యొక్క సరియైన వివరణ II కాదు

2. ✖

Statement-I is true and Statement-II is false

ప్రవచనం-I సత్యము మరియు ప్రవచనం-II అసత్యము

3. ✖

Statement-I is false and Statement-II is true

ప్రవచనం-I అసత్యము మరియు ప్రవచనం-II సత్యము

4. ✔

Maths - Question 35

Original paper page 28; original question and option regions

If each of the coefficients a, b, c in the equation $ax^2 + bx + c = 0$ is determined by throwing a die, then the probability that the equation will have equal roots, is

$ax^2 + bx + c = 0$ సమీకరణంలోని ప్రతి గుణకం a, b, c ని ఒక పాచికను దొర్లించడం ద్వారా నిర్ధారణ చేసేటట్లయితే, ఈ సమీకరణము సమాన మూలములను కలిగి ఉండే సంభావ్యత

Options :

$$\frac{1}{36}$$

1. ✖

Maths - Question 35 (continued)

Original paper page 29; original question and option regions



$$\frac{1}{72}$$



$$\frac{7}{216}$$



$$\frac{5}{216}$$

Maths - Question 37

Original paper page 30; original question and option regions

E_1 and E_2 are two independent events of a random experiment such that $P(E_1) = \frac{1}{2}$ and

$P(E_1 \cup E_2) = \frac{2}{3}$. Then match the items of List-I with the items of List-II.

ఒక యాదృచ్ఛిక ప్రయోగంలో E_1 మరియు E_2 లు $P(E_1) = \frac{1}{2}$ మరియు $P(E_1 \cup E_2) = \frac{2}{3}$

అయ్యేటట్లుగా ఉన్న రెండు స్వతంత్ర ఘటనలు. అప్పుడు జాబితా-I లోని అంశాలను జాబితా-II లోని అంశాలతో జత చేయండి.

	<u>List-I</u> <u>జాబితా-I</u>		<u>List-II</u> <u>జాబితా-II</u>
(A)	$P(E_2)$	(i)	$\frac{1}{2}$
(B)	$P(E_1/E_2)$	(ii)	$\frac{5}{6}$
(C)	$P(\overline{E_2}/E_1)$	(iii)	$\frac{1}{3}$
(D)	$P(\overline{E_1} \cup \overline{E_2})$	(iv)	$\frac{1}{6}$
		(v)	$\frac{2}{3}$

The correct match is

సరియైన జోడి

Maths - Question 37 (continued)

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ons :



A-iii; B-iv; C-i; D-v

A-iii; B-i; C-v; D-ii

A-i; B-v; C-ii; D-iv

A-v; B-i; C-iii; D-ii



Maths - Question 39

Original paper page 32; original question and option regions

If a random variable X has the following probability distribution, then its variance is nearly

$$X = x : \quad -3 \quad -2 \quad -1 \quad 0 \quad 1 \quad 2 \quad 3$$

$$P(X = x): \quad 0.05 \quad 0.1 \quad 2K \quad 0 \quad 0.3 \quad K \quad 0.1$$

ఒక యాదృచ్ఛిక చలరాశి X ఈ క్రింది సంభావ్యతా విభజనాన్ని కలిగి ఉంటే, అప్పుడు దాని విస్తృతి

$$X = x : \quad -3 \quad -2 \quad -1 \quad 0 \quad 1 \quad 2 \quad 3$$

$$P(X = x): \quad 0.05 \quad 0.1 \quad 2K \quad 0 \quad 0.3 \quad K \quad 0.1$$

Options :

2.8875

1. ✓

2.9875

2. ✗

Maths - Question 39 (continued)

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2.7865



2.785

4. ✖

Maths - Question 42

Original paper page 35; original question and option regions



If the axes are rotated through an angle ' α ', then the number of values of α such that the transformed equation of $x^2 + y^2 + 2x + 2y - 5 = 0$ contains no linear terms is

నిరూపక అక్షాలను α కోణంతో భ్రమణం చేస్తే, $x^2 + y^2 + 2x + 2y - 5 = 0$ సమీకరణం యొక్క రూపాంతరం చెందిన సమీకరణంలో ఏక ఘాత పదాలు లుప్తం అయ్యేటట్లుగా ఉండే α యొక్క విలువల సంఖ్య

Options :

1. ✓ 0

2. ✗ 1

3. ✗ 2

4. ✗ Infinite
అనంతం

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Maths - Question 43

Original paper page 35; original question and option regions

A line L passing through the point $P(-5, -4)$ cuts the lines $x - y - 5 = 0$ and $x + 3y + 2 = 0$ respectively at Q and R such that $\frac{18}{PQ} + \frac{15}{PR} = 2$, then slope of the line L is

$P(-5, -4)$ బిందువు గుండా పోయే L అనే రేఖ, $x - y - 5 = 0$ మరియు $x + 3y + 2 = 0$ రేఖలను వరుసగా Q మరియు R ల వద్ద $\frac{18}{PQ} + \frac{15}{PR} = 2$ అయ్యేటట్లు ఖండిస్తే, ఆ రేఖ L యొక్క వాలు

Maths - Question 43 (continued)

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± 1

2. ✖

$\pm \frac{1}{\sqrt{3}}$

3. ✔

$\pm \sqrt{3}$

4. ✖

$\pm \frac{2}{\sqrt{3}}$

am

Maths - Question 45

Original paper page 37; original question and option regions

The equation of a line which makes an angle of 45° with each of the pair of lines

$$xy - x - y + 1 = 0 \text{ is}$$

$xy - x - y + 1 = 0$ రేఖాయుగ్మంలోని ప్రతి రేఖతో 45° కోణాన్ని చేసే ఒక సరళరేఖ సమీకరణం

Options :

1. ✓ $x - y = 5$

2. ✗ $2x + y = 3$

3. ✗ $x + 7y = 8$

4. ✗ $3x - y = 2$

Maths - Question 46

Original paper page 38; original question and option regions

Time : 0



If the slope of one of the lines in the pair of lines $8x^2 + axy + y^2 = 0$ is thrice the slope of the second line, then $a =$

సరళ రేఖాయుగ్మం $8x^2 + axy + y^2 = 0$ లోని ఒక రేఖ వాలు, రెండవ రేఖ వాలునకు మూడు రెట్లు ఉంటే, $a =$

Options :

1. ✓ $8\sqrt{\frac{2}{3}}$

2. ✗ 6

3. ✗ $16\sqrt{2}$

4. ✗ $3\frac{\sqrt{2}}{5}$

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Maths - Question 47

Original paper page 38; original question and option regions

The triangle PQR is inscribed in the circle $x^2 + y^2 = 25$. If $Q = (3, 4)$ and $R = (-4, 3)$ then

$$\angle QPR =$$

త్రిభుజం PQR వృత్తం $x^2 + y^2 = 25$ లో అంతర్లిఖించబడింది. $Q = (3, 4), R = (-4, 3)$

$$\text{అయిన } \angle QPR =$$

Maths - Question 47 (continued)

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$$\frac{\pi}{2}$$

$$\frac{\pi}{3}$$

$$\frac{\pi}{4}$$

$$\frac{\pi}{6}$$

Maths - Question 48

Original paper page 39; original question and option regions

The locus of the point of intersection of perpendicular tangents drawn to the circle $x^2 + y^2 = 10$ is

$x^2 + y^2 = 10$ అనే వృత్తానికి గీసిన లంబస్పర్శరేఖల ఖండన బిందువు యొక్క బిందుపథము

Options :

$x^2 + y^2 = 5$

1. ✖

$x^2 + y^2 = 20$

2. ✔

Maths - Question 48 (continued)

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$$x^2 + y^2 = 25$$

$$x^2 + y^2 = 100$$



Maths - Question 49

Original paper page 40; original question and option regions

The normal drawn at $(1, 1)$ to the circle $x^2 + y^2 - 4x + 6y - 4 = 0$ is

$x^2 + y^2 - 4x + 6y - 4 = 0$ అనే వృత్తం మీద $(1, 1)$ వద్ద గీసిన అభిలంబరేఖ సమీకరణం

Options :

1. ✖

$$4x + 3y = 7$$

2. ✔

$$4x + y = 5$$

3. ✖

$$x + y = 2$$

4. ✖

$$4x - y = 3$$

Parametric equations of the circle $2x^2 + 2y^2 = 9$ are

$2x^2 + 2y^2 = 9$ అనే వృత్తం యొక్క పరామితీయ సమీకరణాలు

Options :

$$x = \frac{3}{2} \cos \theta, \quad y = \frac{3}{2} \sin \theta$$

1. ✖

$$x = \frac{3}{\sqrt{2}} \cos \theta, \quad y = 3 \sin \theta$$

2. ✖

$$x = \frac{3}{\sqrt{2}} \sin \theta, \quad y = \frac{3}{\sqrt{2}} \cos \theta$$

3. ✔

$$x = 3 \sin \theta, \quad y = \frac{3}{2} \cos \theta$$

4. ✖

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Maths - Question 51

Original paper page 41; original question and option regions

Angle between the circles $x^2 + y^2 - 4x - 6y - 3 = 0$ and $x^2 + y^2 + 8x - 4y + 11 = 0$ is

$x^2 + y^2 - 4x - 6y - 3 = 0$ మరియు $x^2 + y^2 + 8x - 4y + 11 = 0$ అనే వృత్తాల మధ్య కోణం

Options :

1. ✓

$$\frac{\pi}{3}$$

Maths - Question 51 (continued)

Original paper page 42; original question and option regions



$$\frac{\pi}{6}$$

$$\frac{\pi}{2}$$

$$\frac{\pi}{4}$$

Maths - Question 52

Original paper page 42; original question and option regions

Equation of the line touching both parabolas $y^2 = 4x$ and $x^2 = -32y$ is

$y^2 = 4x$ మరియు $x^2 = -32y$ పరావలయాల ఉమ్మడి స్పర్శరేఖ సమీకరణం

Options :

1. ✖

$$x + 2y + 4 = 0$$

2. ✖

$$2x + y - 4 = 0$$

3. ✖

$$x - 2y - 4 = 0$$

4. ✔

$$x - 2y + 4 = 0$$

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Maths - Question 52 (continued)

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The length of the latusrectum of $16x^2 + 25y^2 = 400$ is

$16x^2 + 25y^2 = 400$ అనే దీర్ఘవృత్తం యొక్క నాభీలంబము పొడవు

Options :

$$\frac{25}{2}$$

1. ✖

$$\frac{25}{4}$$

2. ✖

$$\frac{16}{2}$$

3. ✖

$$\frac{32}{5}$$

4. ✔

Maths - Question 54

Original paper page 44; original question and option regions



The line $21x + 5y = k$ touches the hyperbola $7x^2 - 5y^2 = 232$ then $k =$

$21x + 5y = k$ అనే రేఖ $7x^2 - 5y^2 = 232$ అనే అతిపరావలయానికి స్పర్శరేఖ అయితే

$k =$

Options :

1. ✓ 116

2. ✗ 232

3. ✗ 58

4. ✗ 110

via.com

Maths - Question 55

Original paper page 44; original question and option regions

If the equation $\frac{x^2}{7-k} + \frac{y^2}{5-k} = 1$ represents a hyperbola, then

$\frac{x^2}{7-k} + \frac{y^2}{5-k} = 1$ అనే సమీకరణం ఒక అతిపరావలయాన్ని సూచిస్తే అప్పుడు,

Options :

$$5 < k < 7$$

1. ✓

$$k < 5 \text{ or } k > 7$$

2. ✗

Maths - Question 55 (continued)

Original paper page 45; original question and option regions



$$k > 5$$

$$k \neq 5, \quad k \neq 7, \quad -\infty < k < \infty$$

Maths - Question 56

Original paper page 45; original question and option regions

If a line L makes angles $\frac{\pi}{3}$ and $\frac{\pi}{4}$ with Y-axis and Z -axis respectively, then the angle between L and another line having direction ratios 1, 1, 1 is

L అనే ఒకరేఖ Y-అక్షం మరియు Z-అక్షంలతో వరుసగా $\frac{\pi}{3}$ మరియు $\frac{\pi}{4}$ కోణాలు చేస్తే, 1, 1, 1 దిక్ నిష్పత్తులు గల మరియుక రేఖతో ఆరేఖ L చేసే కోణం

Options :

$$\cos^{-1}\left(\frac{2}{\sqrt{6}}\right)$$

1. ✖

$$\cos^{-1}\left(\frac{\sqrt{2}+1}{3\sqrt{3}}\right)$$

2. ✖

$$\cos^{-1}\left(\frac{\sqrt{2}-1}{3}\right)$$

3. ✖

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Maths - Question 56 (continued)

Original paper page 46; original question and option regions

$$\cos^{-1} \left(\frac{\sqrt{2}+1}{\sqrt{6}} \right)$$



Maths - Question 57

Original paper page 46; original question and option regions

If l, m, n are the direction cosines of a line that is perpendicular to the lines having the direction ratios $1, 2, -1$ and $1, -2, 1$ then $(l+m+n)^2 =$

1, 2, -1 మరియు 1, -2, 1 లను దిక్ నిష్పత్తులుగా గల రెండు రేఖలకు లంబంగా ఉండే రేఖ యొక్క దిక్ కోసైన్లు l, m, n అయితే, $(l+m+n)^2 =$

Options :

1. ✖ $\frac{1}{20}$

2. ✔ $\frac{9}{5}$

3. ✖ $\frac{1}{5}$

4. ✖ $\frac{3}{20}$

Maths - Question 59

Original paper page 47; original question and option regions

$$\lim_{x \rightarrow \infty} \frac{[2x - 3]}{x} =$$

Options :

0

1. ✖

Maths - Question 59 (continued)

Original paper page 48; original question and option regions



∞

-3

2

Maths - Question 60

Original paper page 48; original question and option regions

$$\lim_{x \rightarrow 0} \frac{\cos 2x - \cos 3x}{\cos 4x - \cos 5x} =$$

Options :

1. ✓ $\frac{5}{9}$

2. ✗ 1

3. ✗ $\frac{3}{4}$

4. ✗ $\frac{2}{5}$

Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Inst
Time : 0

If a real valued function $f(x) = \begin{cases} \frac{2x^2+(k+2)x+9}{3x^2-7x-6}, & \text{for } x \neq 3 \\ l, & \text{for } x=3 \end{cases}$ is continuous at $x=3$ and l is a

finite value, then $l-k=$

ఒక వాస్తవ మూల్యప్రమేయం $f(x) = \begin{cases} \frac{2x^2+(k+2)x+9}{3x^2-7x-6}, & x \neq 3 \\ l, & x=3 \end{cases}$ $x=3$ వద్ద

అవిచ్ఛిన్నము మరియు l ఒక పరిమిత విలువైతే, $l-k=$

Options :

$$\frac{31}{11}$$

1. ✖

$$\frac{124}{11}$$

2. ✔

$$24$$

3. ✖

$$32$$

4. ✖

Maths - Question 62

Original paper page 50; original question and option regions



If $y = \tan^{-1} \frac{x}{1+2x^2} + \tan^{-1} \frac{x}{1+6x^2} + \tan^{-1} \frac{x}{1+12x^2}$, then $\left(\frac{dy}{dx}\right)_{x=\frac{1}{2}} =$

$y = \tan^{-1} \frac{x}{1+2x^2} + \tan^{-1} \frac{x}{1+6x^2} + \tan^{-1} \frac{x}{1+12x^2}$ అయితే, $\left(\frac{dy}{dx}\right)_{x=\frac{1}{2}} =$

Options :

1. ✖ 1

2. ✖ -1

3. ✔ 0

4. ✖ $\frac{1}{2}$

ndia.com

Maths - Question 63

Original paper page 50; original question and option regions

If $f(x) = 5 \cos^3 x - 3 \sin^2 x$ and $g(x) = 4 \sin^3 x + \cos^2 x$, then the derivative of $f(x)$ with respect to $g(x)$ is

$f(x) = 5 \cos^3 x - 3 \sin^2 x$ మరియు $g(x) = 4 \sin^3 x + \cos^2 x$ అయితే, $g(x)$ ద్వారా $f(x)$ యొక్క అవకలజం

Options :

$$\frac{5 \cos x + 2}{6 \cos x - 1}$$

1. ✖

Maths - Question 63 (continued)

Original paper page 51; original question and option regions



$$-\left(\frac{5 \cos x + 2}{6 \cos x - 1}\right)$$

$$\frac{15 \cos x - 6}{12 \sin x + 2}$$

$$-\left(\frac{15 \cos x + 6}{12 \sin x - 2}\right)$$

im

Maths - Question 64

Original paper page 51; original question and option regions

If $y = 1 + x + x^2 + x^3 + \dots \infty$ and $|x| < 1$, then $y'' =$

$y = 1 + x + x^2 + x^3 + \dots \infty$ మరియు $|x| < 1$ అయితే, $y'' =$

Options :

1. ✓

$$2yy'$$

2. ✗

$$\frac{2y}{y'}$$

3. ✗

$$\frac{y'}{2y}$$

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Maths - Question 64 (continued)

Original paper page 52; original question and option regions



$$2y^2y'$$

Maths - Question 67

Original paper page 53; original question and option regions

If the curves $2x^2 + ky^2 = 30$ and $3y^2 = 28x$ cut each other orthogonally, then $k =$

$2x^2 + ky^2 = 30, 3y^2 = 28x$ వక్రాలు పరస్పరం లంబంగా ఖండించుకుంటే, $k =$

Maths - Question 67 (continued)

Original paper page 54; original question and option regions

5

3

2

1

Maths - Question 68

Original paper page 54; original question and option regions

The interval containing all the real values of x such that the real valued function $f(x) = \sqrt{x} + \frac{1}{\sqrt{x}}$ is strictly increasing is

వాస్తవ మూల్య ప్రమేయం $f(x) = \sqrt{x} + \frac{1}{\sqrt{x}}$ శుద్ధ ఆరోహణ ప్రమేయమయ్యేటట్లుగా ఉండే x యొక్క వాస్తవ విలువలన్నింటిని కలిగిన అంతరం

Options :

1. ✓ $(1, \infty)$

2. ✗ $(0, 1)$

3. ✗ $(-\infty, 0) \cup (1, \infty)$

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Maths - Question 68 (continued)

Original paper page 55; original question and option regions

$$(-\infty, 0)$$



Maths - Question 69

Original paper page 55; original question and option regions

$$\int e^{4x^2+8x-4} (x+1) \cos(3x^2+6x-4) \, dx =$$

Options :

$$\frac{e^{4x^2+8x-4}}{25} [3 \sin(3x^2+6x-4) - 4 \cos(3x^2+6x-4)] + c$$

1. ✖

$$\frac{e^{4x^2+8x-4}}{50} [4 \cos(3x^2+6x-4) + 3 \sin(3x^2+6x-4)] + c$$

2. ✔

$$\frac{e^{4x^2+8x-4}}{25} [3 \cos(3x^2+6x-4) + 4 \sin(3x^2+6x-4)] + c$$

3. ✖

$$\frac{e^{4x^2+8x-4}}{50} [4 \sin(3x^2+6x-4) - 3 \cos(3x^2+6x-4)] + c$$

4. ✖

Maths - Question 70

Original paper page 56; original question and option regions

Time : 0



$$\int [(\log 2x)^2 + 2 \log 2x] dx =$$

Options :

1. ✖ $(\log 2x)^2 + c$

2. ✖ $2x \log 2x + c$

3. ✔ $x(\log 2x)^2 + c$

4. ✖ $2x(\log x)^2 + c$

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Maths - Question 71

Original paper page 56; original question and option regions

If $\int \log(6 \sin^2 x + 17 \sin x + 12)^{\cos x} dx = f(x) + c$ then, $f\left(\frac{\pi}{2}\right) =$

$\int \log(6 \sin^2 x + 17 \sin x + 12)^{\cos x} dx = f(x) + c$ అయితే, $f\left(\frac{\pi}{2}\right) =$

Options :

$$\frac{1}{6} [\log 5^5 + \log 7^7 - 12]$$

1. ✖

Maths - Question 71 (continued)

Original paper page 57; original question and option regions



$$\frac{1}{6}[7 \log 5 + 5 \log 7 + 29]$$

3. ✖

$$\frac{1}{6}[14 \log 5 + 15 \log 7 + 12]$$

4. ✔

$$\frac{1}{6}[15 \log 5 + 14 \log 7 - 29]$$

Maths - Question 72

Original paper page 57; original question and option regions

$$\int \frac{1}{(1+x^2)\sqrt{x^2+2}} dx =$$

Options :

1. ✓

$$- \operatorname{Tan}^{-1} \frac{\sqrt{x^2+2}}{|x|} + c$$

2. ✗

$$- \operatorname{Tan}^{-1} \sqrt{x^2+2} + c$$

3. ✗

$$\operatorname{Tan}^{-1} \sqrt{\frac{x^2+1}{x^2+2}} + c$$

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Maths - Question 72 (continued)

Original paper page 58; original question and option regions

$$- \operatorname{Tan}^{-1} \sqrt{\frac{x^2 + 2}{x^2 + 1}} + c$$



$$\int \sin^4 x \cos^4 x \, dx =$$

Options :

$$\frac{1}{128}(-2 \sin^3 x \cos x - 3 \sin x \cos x + 3) + c$$

1. ✖

$$\frac{1}{256}(-2 \sin^3 2x \cos 2x - 3 \sin 2x \cos 2x + 6x) + c$$

2. ✔

$$\frac{1}{128}(2 \sin^3 x \cos x - 3 \sin x \cos x + 3x) + c$$

3. ✖

$$\frac{1}{256}(3 \sin^3 x \cos x - 2 \sin x \cos x + 2) + c$$

4. ✖

Maths - Question 74

Original paper page 59; original question and option regions

Time : 0



$$\int_0^1 \sqrt{\frac{2+x}{2-x}} dx =$$

Options :

1. ✖ $\pi+2$

2. ✖ $\frac{1}{2}(\pi+2)$

3. ✖ $\frac{\pi}{2}+2+\sqrt{3}$

4. ✔ $\frac{\pi}{3}+2-\sqrt{3}$

india.com

Maths - Question 75

Original paper page 59; original question and option regions

If $M = \int_0^{\infty} \frac{\log t}{1+t^3} dt$ and $N = \int_{-\infty}^{\infty} \frac{e^{2t} t}{1+e^{3t}} dt$, then

$M = \int_0^{\infty} \frac{\log t}{1+t^3} dt$ మరియు $N = \int_{-\infty}^{\infty} \frac{e^{2t} t}{1+e^{3t}} dt$ అయితే

Options :

$$N = 2M$$

1. ✖

Maths - Question 75 (continued)

Original paper page 60; original question and option regions



$$N = M$$

$$N = 3M$$

$$N = -M$$

Maths - Question 76

Original paper page 60; original question and option regions

$$\int_{-2}^2 (4 - x^2)^{\frac{5}{2}} dx =$$

Options :

1. ✖ 40π

2. ✔ 20π

3. ✖ 10π

4. ✖ $\frac{5\pi}{32}$

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Maths - Question 76 (continued)

Original paper page 61; original question and option regions



Maths - Question 77

Original paper page 61; original question and option regions

$$\lim_{x \rightarrow \infty} \left[\left(1 + \frac{1}{n^3}\right)^{\frac{1}{n^3}} \left(1 + \frac{8}{n^3}\right)^{\frac{4}{n^3}} \left(1 + \frac{27}{n^3}\right)^{\frac{9}{n^3}} \dots (2)^{\frac{1}{n}} \right] =$$

Options :

$$\log 2 - \frac{1}{2}$$

1. ✖

$$e^{\left(\log 2 - \frac{1}{2}\right)}$$

2. ✖

$$e^{\left(\frac{2\log 2 - 1}{3}\right)}$$

3. ✔

$$\frac{1}{3}(2\log 2 - 1)$$

4. ✖

Maths - Question 78

Original paper page 61; original question and option regions

$$\int_{-5\pi}^{5\pi} (1 - \cos 2x)^{\frac{5}{2}} dx =$$

Maths - Question 78 (continued)

Original paper page 62; original question and option regions

$$\frac{64\sqrt{2}}{5}$$

$$\frac{128\sqrt{2}}{5}$$

$$\frac{256\sqrt{2}}{3}$$

$$\frac{128\sqrt{2}}{3}$$

The general solution of the differential equation $(xy + y^2)dx - (x^2 - 2xy)dy = 0$ is

$(xy + y^2)dx - (x^2 - 2xy)dy = 0$ అనే అవకలన సమీకరణం యొక్క సాధారణ సాధన

Options :

1. ✖ $c x y^2 = e^{x/y}$

2. ✔

$c x y^2 e^{x/y} = 1$

3. ✖

$c x y e^{x/y} = 1$

4. ✖

$c x y = e^{x/y}$

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Maths - Question 80 (continued)

Original paper page 64; original question and option regions



Maths - Question 1

Original paper page 2; original question and option regions

If $A \subseteq \mathbb{Z}$ and the function $f: A \rightarrow \mathbb{R}$ is defined by $f(x) = \frac{1}{\sqrt{64 - (0.5)^{24+x-x^2}}}$ then the sum of all

absolute values of elements of A is

$A \subseteq \mathbb{Z}$ మరియు $f: A \rightarrow \mathbb{R}$ ప్రమేయం $f(x) = \frac{1}{\sqrt{64 - (0.5)^{24+x-x^2}}}$ అయ్యేట్లు నిర్వచితమైనచో

A నందలి మూలకాలన్నింటి పరమ మూల్య విలువల మొత్తము

Options :

36

1. ✖

Maths - Question 1 (continued)

Original paper page 3; original question and option regions

5

25

11



Which of the following functions are odd?

ఈ క్రింది ప్రమేయాలలో ఏవి బేసి ప్రమేయాలు

I. $f(x) = x \left(\frac{e^x - 1}{e^x + 1} \right)$

II. $f(x) = k^x + k^{-x} + \cos x$

III. $f(x) = \log(x + \sqrt{x^2 + 1})$

Options :

1. ✖

II

2. ✖

I, II

3. ✔

III

4. ✖

I

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Maths - Question 2 (continued)

Original paper page 4; original question and option regions



Maths - Question 3

Original paper page 4; original question and option regions

The n^{th} term of the series $1 + (3 + 5 + 7) + (9 + 11 + 13 + 15 + 17) + \dots$

$1 + (3 + 5 + 7) + (9 + 11 + 13 + 15 + 17) + \dots$ శ్రేణి యొక్క n వ పదము

Options :

$$(2n + 1) [n^2 - (n - 1)^2]$$

1. ✖

$$(2n - 1) [(n - 1)^2 - n^2]$$

2. ✖

$$(2n + 1) [(n - 1)^2 - n^2]$$

3. ✖

$$(2n - 1) [(n - 1)^2 + n^2]$$

4. ✔

Maths - Question 4

Original paper page 5; original question and option regions



If $A = \begin{bmatrix} 83 & 74 & 41 \\ 93 & 96 & 31 \\ 24 & 15 & 79 \end{bmatrix}$, then $\det(A - A^T) =$

$A = \begin{bmatrix} 83 & 74 & 41 \\ 93 & 96 & 31 \\ 24 & 15 & 79 \end{bmatrix}$ అయితే, $\det(A - A^T) =$

Options :

1. ✓ 0

2. ✗ -7851

3. ✗ 2442

4. ✗ 1

Maths - Question 5

Original paper page 5; original question and option regions

$$\text{If } \begin{vmatrix} a & 1 & 1 \\ 1 & b & 1 \\ 1 & 1 & c \end{vmatrix} > 0, \text{ then } abc >$$

$$\begin{vmatrix} a & 1 & 1 \\ 1 & b & 1 \\ 1 & 1 & c \end{vmatrix} > 0 \text{ అయితే, } abc >$$

Maths - Question 5 (continued)

Original paper page 6; original question and option regions

ons :



1

−8

8

3

Maths - Question 6

Original paper page 6; original question and option regions

If the system of equations $a_1x + b_1y + c_1z = 0$, $a_2x + b_2y + c_2z = 0$, $a_3x + b_3y + c_3z = 0$ has only

trivial solution, then the rank of $\begin{bmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{bmatrix}$ is

సమీకరణ వ్యవస్థ $a_1x + b_1y + c_1z = 0$, $a_2x + b_2y + c_2z = 0$, $a_3x + b_3y + c_3z = 0$, త్రివిధసాధనను

మాత్రమే కలిగి ఉంటే, మాత్రిక $\begin{bmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{bmatrix}$ యొక్క ర్యాంక్

Options :

1. ✖ 2

2. ✖ 1

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Maths - Question 6 (continued)

Original paper page 7; original question and option regions

3

0



ω is a complex cube root of unity and if Z is a complex number satisfying $|Z - 1| \leq 2$ and

$|\omega^2 Z - 1 - \omega| = a$, then the set of possible values of a is

ఏకకం యొక్క ఘన మూలాలలో ω ఒక సంకీర్ణ మూలము, $|Z - 1| \leq 2$ మరియు

$|\omega^2 Z - 1 - \omega| = a$, లను తృప్తి పరిచే ఒక సంకీర్ణ సంఖ్య Z అయితే, అప్పుడు a కి

సాధ్యమయ్యే విలువల సమితి

Options :

$$0 \leq a \leq 2$$

1. ✖

$$\frac{1}{2} \leq a \leq \frac{\sqrt{3}}{2}$$

2. ✖

$$|\omega| \leq a \leq \frac{\sqrt{3}}{2} + 2$$

3. ✖

$$0 \leq a \leq 4$$

4. ✔

Question Number : 8 Question Id : 4509387688 Display Question Number :  **Yes Is Quest**

Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Inst
Time : 0

If the roots of the equation $Z^3 + iZ^2 + 2i = 0$ are the vertices of a triangle ABC, then that triangle ABC is

$Z^3 + iZ^2 + 2i = 0$ సమీకరణం యొక్క మూలాలు ఒక త్రిభుజం ABC యొక్క శీర్షములు అయితే ఆ త్రిభుజము ABC ఒక

Options :

a right angled triangle

లంబకోణ త్రిభుజము

1. ✖

an equilateral triangle

సమబాహు త్రిభుజము

2. ✖

an isosceles triangle

సమద్విబాహు త్రిభుజము

3. ✔

a right angled isosceles triangle

లంబకోణ సమద్విబాహు త్రిభుజము

4. ✖

Maths - Question 9

Original paper page 9; original question and option regions



(r, θ) denotes $r(\cos \theta + i \sin \theta)$. If $x = (1, \alpha), y = (1, \beta), z = (1, \gamma)$ and $x + y + z = 0$ then

$$\sum \cos(2\alpha - \beta - \gamma) =$$

(r, θ) అనేది $r(\cos \theta + i \sin \theta)$ ను సూచిస్తుంది. $x = (1, \alpha), y = (1, \beta), z = (1, \gamma)$ మరియు

$$x + y + z = 0$$
 అయితే $\sum \cos(2\alpha - \beta - \gamma) =$

Options :

1. ✓ 3

2. ✗ 0

3. ✗ 1

4. ✗ -1

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Maths - Question 10

Original paper page 9; original question and option regions

The set of all real values of x satisfying the inequality $\frac{7x^2-5x-18}{2x^2+x-6} < 2$ is

$\frac{7x^2-5x-18}{2x^2+x-6} < 2$ అసమీకరణాన్ని తృప్తి పరచే x యొక్క అన్ని వాస్తవ విలువల సమితి

 Options :

$$\left(-\infty, -\frac{2}{3}\right] \cup [3, \infty)$$

1. ✖

Maths - Question 10 (continued)

Original paper page 10; original question and option regions



$$\left(-2, -\frac{2}{3}\right) \cup \left(\frac{3}{2}, 3\right)$$

$$(-\infty, -2) \cup \left(\frac{3}{2}, \infty\right)$$

3. ✖

$$\left[-\frac{2}{3}, \frac{3}{2}\right)$$

4. ✖

Maths - Question 11

Original paper page 10; original question and option regions

The set of all values of k for which the inequality $x^2 - (3k+1)x + 4k^2 + 3k - 3 > 0$ is true for all real values of x , is

x యొక్క అన్ని వాస్తవ విలువలకు $x^2 - (3k+1)x + 4k^2 + 3k - 3 > 0$ సత్యమయ్యేటట్లు ఉండే k యొక్క అన్ని విలువల సమితి

Options :

1. ✖ $\left(-\frac{13}{7}, 1\right)$

2. ✖ $\left(-1, \frac{13}{7}\right)$

Maths - Question 11 (continued)

Original paper page 11; original question and option regions



$$\left(-\infty, -\frac{13}{7}\right) \cup (1, \infty)$$

$$(-\infty, -1) \cup \left(\frac{13}{7}, \infty\right)$$

4. ✖

Maths - Question 12

Original paper page 11; original question and option regions

The cubic equation whose roots are the squares of the roots of the equation

$$12x^3 - 20x^2 + x + 3 = 0 \text{ is}$$

$12x^3 - 20x^2 + x + 3 = 0$ సమీకరణం యొక్క మూలాల వర్గాలను మూలాలుగా గలిగిన ఘన సమీకరణం

Options :

$$x^3 + 376x^2 - 121x - 9 = 0$$

1. ✖

$$144x^3 - 400x^2 + 121x + 98 = 0$$

2. ✖

$$144x^3 - 376x^2 + 121x - 9 = 0$$

3. ✔

$$x^3 + 400x^2 - 121x - 98 = 0$$

4. ✖

Question Number : 13 Question Id : 4509387693 Display Question Number :  **Yes Is Que**

Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Inst
Time : 0

α, β, γ are the roots of the equation $x^3 + 3x^2 - 10x - 24 = 0$. If $\alpha(\beta + \gamma)$, $\beta(\gamma + \alpha)$ and $\gamma(\alpha + \beta)$ are the roots of the equation $x^3 + px^2 + qx + r = 0$, then $q =$

$x^3 + 3x^2 - 10x - 24 = 0$ సమీకరణం యొక్క మూలాలు α, β, γ . $\alpha(\beta + \gamma), \beta(\gamma + \alpha)$

మరియు $\gamma(\alpha + \beta)$ లు $x^3 + px^2 + qx + r = 0$ సమీకరణం యొక్క మూలాలైతే, $q =$

Options :

1. ✖ - 44

2. ✖ - 28

3. ✖ 44

4. ✔ 28

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Maths - Question 14

Original paper page 12; original question and option regions

Among the 4 digit numbers formed using the digits 0, 1, 2, 3, 4 when repetition of digits allowed, the number of numbers which are divisible by 4 is

0, 1, 2, 3, 4 అంకెలను వాడుతూ, అంకెల పునరావృతం ను అనుమతించబడినప్పుడు, ఏర్పరచగలిగే 4 అంకెల సంఖ్యలలో, 4 చే భాగింపబడే సంఖ్యల సంఖ్య

Maths - Question 14 (continued)

Original paper page 13; original question and option regions



140

2. ✓ 160

3. ✗ 180

4. ✗ 200

Maths - Question 15

Original paper page 13; original question and option regions

The number of ways of arranging 2 red, 3 white and 5 yellow roses of different sizes into a garland such that no two yellow roses come together is

అన్నీ విభిన్న పరిమాణం గలిగిన 2 ఎర్రని, 3 తెల్లని మరియు 5 పసుపు పచ్చని గులాబీ పువ్వులను, ఏ రెండు పసుపు పచ్చని గులాబీ పువ్వులూ పక్క పక్కన లేకుండా ఉండేటట్లుగా ఒక దండగా కూర్చుంటే విధాల సంఖ్య

Options :

1. ✖ 2880

2. ✖ 144

3. ✔ 1440

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Maths - Question 15 (continued)

Original paper page 14; original question and option regions

288



Maths - Question 17

Original paper page 14; original question and option regions

The independent term in the expansion of $(1 + x + 2x^2) \left(\frac{3x^2}{2} - \frac{1}{3x} \right)^9$ is

$(1 + x + 2x^2) \left(\frac{3x^2}{2} - \frac{1}{3x} \right)^9$ యొక్క విస్తరణలో స్వతంత్రపదము

Maths - Question 17 (continued)

Original paper page 15; original question and option regions

ons :



$$\frac{18}{7}$$

$$\frac{7}{18}$$

$$-\frac{7}{18}$$

$$-\frac{18}{7}$$

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Maths - Question 18

Original paper page 15; original question and option regions

For $|x| < \frac{1}{\sqrt{2}}$, the coefficient of x in the expansion of $\frac{(1-4x)^2(1-2x^2)^{1/2}}{(4-x)^{3/2}}$ is

$|x| < \frac{1}{\sqrt{2}}$ కి, $\frac{(1-4x)^2(1-2x^2)^{1/2}}{(4-x)^{3/2}}$ యొక్క విస్తరణలో x యొక్క గుణకం

Options :

$$\frac{61}{64}$$

1. ✖

$$-\frac{61}{64}$$

2. ✔

Maths - Question 18 (continued)

Original paper page 16; original question and option regions

$$\frac{69}{64}$$

$$-\frac{69}{64}$$



Maths - Question 19

Original paper page 16; original question and option regions

$$\frac{4x^2+5}{(x-2)^4} = \frac{A}{(x-2)} + \frac{B}{(x-2)^2} + \frac{C}{(x-2)^3} + \frac{D}{(x-2)^4} \text{ then } \sqrt{\frac{A}{C} + \frac{B}{C} + \frac{D}{C}} =$$

$$\frac{4x^2+5}{(x-2)^4} = \frac{A}{(x-2)} + \frac{B}{(x-2)^2} + \frac{C}{(x-2)^3} + \frac{D}{(x-2)^4} \text{ అయితే } \sqrt{\frac{A}{C} + \frac{B}{C} + \frac{D}{C}} =$$

Options :

1. ✖

$$\frac{\sqrt{29}}{4}$$

2. ✖

$$\frac{\sqrt{23}}{4}$$

3. ✔

$$\frac{5}{4}$$

4. ✖

$$\frac{4}{5}$$

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Maths - Question 19 (continued)

Original paper page 17; original question and option regions



Maths - Question 20

Original paper page 17; original question and option regions

$$\tan^2 \frac{\pi}{16} + \tan^2 \frac{2\pi}{16} + \tan^2 \frac{3\pi}{16} + \tan^2 \frac{4\pi}{16} + \tan^2 \frac{5\pi}{16} + \tan^2 \frac{6\pi}{16} + \tan^2 \frac{7\pi}{16} =$$

Options :

1. ✓ 35

2. ✗ 41

3. ✗ 37

4. ✗ 33

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Maths - Question 21

Original paper page 17; original question and option regions

$$\sin^2 18^\circ + \sin^2 24^\circ + \sin^2 36^\circ + \sin^2 42^\circ + \sin^2 78^\circ + \sin^2 90^\circ + \sin^2 96^\circ + \sin^2 102^\circ \\ + \sin^2 138^\circ + \sin^2 162^\circ =$$

Options :

1.  $\frac{11}{2}$

Maths - Question 21 (continued)

Original paper page 18; original question and option regions



$$\frac{9}{2}$$

5

4

Maths - Question 22

Original paper page 18; original question and option regions

If A, B, C are the angles of a triangle, then $\frac{\sin A + \sin B + \sin C}{\sin^2 \frac{A}{2} - \sin^2 \frac{B}{2} + \sin^2 \frac{C}{2} - 1} =$

A, B, C లు త్రిభుజ కోణాలైతే, $\frac{\sin A + \sin B + \sin C}{\sin^2 \frac{A}{2} - \sin^2 \frac{B}{2} + \sin^2 \frac{C}{2} - 1} =$

Options :

1. ✖ $-2 \tan \frac{B}{2}$

2. ✔ $-2 \cot \frac{B}{2}$

3. ✖ $2 \tan \frac{B}{2}$

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Maths - Question 22 (continued)

Original paper page 19; original question and option regions

$$2 \cot \frac{B}{2}$$



The general solution of $\cot \frac{x}{2} - \cot x = \operatorname{cosec} \frac{x}{2}$ is

$$\cot \frac{x}{2} - \cot x = \operatorname{cosec} \frac{x}{2} \text{ సమీకరణం యొక్క సాధారణ సాధన}$$

Options :

$$\left\{ 2n\pi + \frac{\pi}{3} \mid n \in \mathbb{Z} \right\}$$

1. ✖

$$\left\{ 4n\pi + \frac{\pi}{3} \mid n \in \mathbb{Z} \right\}$$

2. ✖

$$\left\{ 2n\pi + \frac{2\pi}{3} \mid n \in \mathbb{Z} \right\}$$

3. ✖

$$\left\{ 4n\pi \pm \frac{2\pi}{3} \mid n \in \mathbb{Z} \right\}$$

4. ✔

Maths - Question 24

Original paper page 20; original question and option regions



If $0 < x < \frac{1}{2}$ and $\alpha = \sin^{-1} x + \cos^{-1} \left(\frac{x}{2} + \frac{\sqrt{3-3x^2}}{2} \right)$, then $\tan \alpha + \cot \alpha =$

$0 < x < \frac{1}{2}$ మరియు $\alpha = \sin^{-1} x + \cos^{-1} \left(\frac{x}{2} + \frac{\sqrt{3-3x^2}}{2} \right)$ అయితే, $\tan \alpha + \cot \alpha =$

Options :

1. ✓ $\frac{4}{\sqrt{3}}$

2. ✗ $4\sqrt{3}$

3. ✗ $\frac{4x}{1-x^2}$

4. ✗ $x\sqrt{1-x^2}$

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Maths - Question 25

Original paper page 20; original question and option regions

$$\cosh(\log 4) =$$

Options :

$$\frac{8}{17}$$

1. ✖

Maths - Question 25 (continued)

Original paper page 21; original question and option regions



$$\frac{17}{8}$$

$$0$$

$$\frac{9}{8}$$

Maths - Question 26

Original paper page 21; original question and option regions

$$\text{In } \triangle ABC, a^2 \sin 2B + b^2 \sin 2A =$$

$$\triangle ABC \text{ } \mathcal{S}, a^2 \sin 2B + b^2 \sin 2A =$$

Options :

1. ✖ $2ab \cos A$

2. ✖ $2ab \sin A$

3. ✔ $2ab \sin C$

4. ✖ $2ab \cos C$

Maths - Question 27

Original paper page 22; original question and option regions

Question Number : 27 Question Id : 4509387707 Display Question Number : Yes Is Question

Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Inst

Time : 0

$$\text{In } \triangle ABC, \frac{r_2(r_1+r_3)}{\sqrt{r_1r_2+r_2r_3+r_3r_1}} =$$

$$\triangle ABC \text{ } \mathcal{S}^{\delta} \frac{r_2(r_1+r_3)}{\sqrt{r_1r_2+r_2r_3+r_3r_1}} =$$

Options :

1. ✘ a

2. ✔ b

3. ✘ c

4. ✘ s

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Maths - Question 28

Original paper page 22; original question and option regions

$$\text{In } \triangle ABC, (r_2 + r_3) \operatorname{cosec}^2 \frac{A}{2} =$$

$$\triangle ABC \mathcal{S}^{\delta}, (r_2 + r_3) \operatorname{cosec}^2 \frac{A}{2} =$$

Options :

1. ✖ $4R$

Maths - Question 28 (continued)

Original paper page 23; original question and option regions



$$4R \cot^2 \frac{A}{2}$$

3. ✖

$$4R \tan^2 \frac{A}{2}$$

4. ✖

$$R \tan^2 \frac{A}{2}$$

Maths - Question 29

Original paper page 23; original question and option regions

If the vectors $a\bar{i} + \bar{j} + \bar{k}$; $\bar{i} + b\bar{j} + \bar{k}$; $\bar{i} + \bar{j} + c\bar{k}$ ($a \neq b \neq c \neq 1$) are coplanar, then

$$\frac{1}{1-a} + \frac{1}{1-b} + \frac{1}{1-c} =$$

$a\bar{i} + \bar{j} + \bar{k}$; $\bar{i} + b\bar{j} + \bar{k}$; $\bar{i} + \bar{j} + c\bar{k}$, ($a \neq b \neq c \neq 1$) అనే సదిశలు సతలీయాలైతే,

$$\frac{1}{1-a} + \frac{1}{1-b} + \frac{1}{1-c} =$$

Options :

1. ✖ 0

2. ✔ 1

3. ✖ 2

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Maths - Question 29 (continued)

Original paper page 24; original question and option regions

—1



Maths - Question 30

Original paper page 24; original question and option regions

If $\overline{AB} = 2\bar{i} + 3\bar{j} - 6\bar{k}$; $\overline{BC} = 6\bar{i} - 2\bar{j} + 3\bar{k}$ are the vectors along two sides of a triangle ABC, then perimeter of triangle ABC is:

$\overline{AB} = 2\bar{i} + 3\bar{j} - 6\bar{k}$; $\overline{BC} = 6\bar{i} - 2\bar{j} + 3\bar{k}$ అనే సదిశలు ABC త్రిభుజంలోని రెండు భుజాల వెంబడి ఉన్న సదిశలైతే, త్రిభుజం ABC చుట్టుకొలత

Options :

1. ✖ 21

2. ✔ $\sqrt{74} + 14$

3. ✖ $\sqrt{74} + 19$

4. ✖ $\sqrt{74} + 3$

Maths - Question 31

Original paper page 25; original question and option regions



The orthogonal projection vector of $\vec{a} = 2\vec{i} + 3\vec{j} + 3\vec{k}$ on $\vec{b} = \vec{i} - 2\vec{j} + \vec{k}$ is

$\vec{b} = \vec{i} - 2\vec{j} + \vec{k}$ పై $\vec{a} = 2\vec{i} + 3\vec{j} + 3\vec{k}$ యొక్క లంబ విక్షేప సదిశ

Options :

$$-\frac{1}{6}(2\vec{i} + 3\vec{j} + 3\vec{k})$$

1. ✖

$$\frac{1}{6}(-\vec{i} + 2\vec{j} - \vec{k})$$

2. ✔

$$\vec{i} - 2\vec{j} + \vec{k}$$

3. ✖

$$-\vec{i} + 2\vec{j} - \vec{k}$$

4. ✖

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Maths - Question 32

Original paper page 25; original question and option regions

If $\vec{a} = -4\vec{i} + 2\vec{j} + 4\vec{k}$, $\vec{b} = \sqrt{2}\vec{i} - \sqrt{2}\vec{j}$ are two vectors then angle between the vectors $2\vec{a}$ and $\frac{\vec{b}}{2}$ is

$\vec{a} = -4\vec{i} + 2\vec{j} + 4\vec{k}$, $\vec{b} = \sqrt{2}\vec{i} - \sqrt{2}\vec{j}$ లు రెండు సదిశలైతే, $2\vec{a}, \frac{\vec{b}}{2}$ సదిశల మధ్య కోణం

Options :

30°

1. ✖

Maths - Question 32 (continued)

Original paper page 26; original question and option regions

135°

90°

0°



Maths - Question 33

Original paper page 26; original question and option regions

A unit vector perpendicular to the vectors $\vec{a} = 2\vec{i} + 3\vec{j} + 4\vec{k}$ and $\vec{b} = 3\vec{j} + 2\vec{k}$ is

$\vec{a} = 2\vec{i} + 3\vec{j} + 4\vec{k}$ మరియు $\vec{b} = 3\vec{j} + 2\vec{k}$ సదిశలకు లంబంగా ఉండే ఒక సదిశ

Options :

$$\frac{3\vec{i} + 2\vec{j} - 2\vec{k}}{\sqrt{22}}$$

1. ✖

$$\frac{3\vec{i} + 2\vec{j} - 3\vec{k}}{\sqrt{22}}$$

2. ✔

$$\frac{3\vec{i} - 2\vec{j} + 3\vec{k}}{\sqrt{22}}$$

3. ✖

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Maths - Question 33 (continued)

Original paper page 27; original question and option regions

$$\frac{3\bar{i} + 2\bar{j} + 3\bar{k}}{\sqrt{22}}$$



Maths - Question 36

Original paper page 28; original question and option regions

It is given that in a random experiment events A and B are such that $P(A) = \frac{1}{4}$,

$$P(A|B) = \frac{1}{2} \text{ and } P(B|A) = \frac{2}{3} \text{ then } P(B) =$$

ఒక యాదృచ్ఛిక ప్రయోగంలో A మరియు B ఘటనలు $P(A) = \frac{1}{4}$, $P(A|B) = \frac{1}{2}$, $P(B|A) =$

$$\frac{2}{3} \text{ అయ్యేటట్లుగా ఉంటే, } P(B) =$$

Maths - Question 36 (continued)

Original paper page 29; original question and option regions

$$\frac{1}{3}$$

$$\frac{2}{3}$$

$$\frac{1}{2}$$

$$\frac{1}{6}$$

Maths - Question 39

Original paper page 31; original question and option regions

If the probability distribution of a random variable X is as follows, then $k =$

$X = x$	1	2	3	4
$P(X = x)$	$2k$	$4k$	$3k$	k

ఒక యాదృచ్ఛిక చలరాశి X యొక్క సంభావ్యత విభాజనం క్రింది విధంగా ఉంటే, $k =$

$X = x$	1	2	3	4
$P(X = x)$	$2k$	$4k$	$3k$	k

Options :

1. ✓

$$\frac{1}{10}$$

2. ✗

$$\frac{2}{10}$$

3. ✗

$$\frac{3}{10}$$

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Maths - Question 39 (continued)

Original paper page 32; original question and option regions



$$\frac{4}{10}$$

The origin is shifted to the point $(2, 3)$ by translation of axes and then the coordinate axes are rotated about the origin through an angle θ in the counter-clockwise sense. Due to this if the equation $3x^2 + 2xy + 3y^2 - 18x - 22y + 50 = 0$ is transformed to $4x^2 + 2y^2 - 1 = 0$, then the angle $\theta =$

సమాంతర అక్షపరివర్తన ద్వారా మూలబిందువును $(2, 3)$ బిందువుకు మార్చి ఆ తరువాత నిరూపక అక్షాలను మూలబిందువు పరంగా ధనదిశలో θ కోణంతో భ్రమణం చెందించారు. దీనివల్ల $3x^2 + 2xy + 3y^2 - 18x - 22y + 50 = 0$ అనే సమీకరణం $4x^2 + 2y^2 - 1 = 0$ అనే సమీకరణంగా రూపాంతరం చెందితే, అప్పుడు కోణం $\theta =$

Options :

1. ✓ $\frac{\pi}{4}$

2. ✗ $\frac{\pi}{3}$

3. ✗ $\frac{\pi}{6}$

4. ✗ $\frac{\pi}{2}$

Maths - Question 43

Original paper page 35; original question and option regions



If the straight line passing through P (3, 4) makes an angle $\frac{\pi}{6}$ with the positive x-axis in anticlockwise direction and meets the line $12x + 5y + 10 = 0$ at Q, then the length of the segment PQ is

P (3, 4) గుండా పోయే సరళరేఖ ధన x-అక్షంతో ధనదిశలో కోణం $\frac{\pi}{6}$ చేస్తూ మరియు రేఖ

$12x + 5y + 10 = 0$ ను Q వద్ద కలుస్తుంటే, అప్పుడు రేఖాఖండము PQ యొక్క పొడవు

Options :

$$\frac{64}{12\sqrt{2} + 1}$$

1. ✖

$$\frac{96}{9\sqrt{2} - 1}$$

2. ✖

$$\frac{112}{10\sqrt{3} + 3}$$

3. ✖

$$\frac{132}{12\sqrt{3} + 5}$$

4. ✔

Maths - Question 44

Original paper page 36; original question and option regions



The equations of the perpendicular bisectors of the sides AB and AC of ΔABC are $x - y + 5 = 0$

and $x + 2y = 0$ respectively. If the coordinates of A are $(1, -2)$, then the equation of the line BC is

ΔABC యొక్క భుజములు AB మరియు AC ల లంబసమద్విఖండన రేఖల సమీకరణాలు

వరుసగా $x - y + 5 = 0$ మరియు $x + 2y = 0$. A యొక్క నిరూపకాలు $(1, -2)$ అయితే, రేఖ BC

యొక్క సమీకరణము

Options :

$$14x + 23y - 40 = 0$$

1. ✓

$$13x - 9y - 14 = 0$$

2. ✗

$$9x - 14y - 25 = 0$$

3. ✗

$$8x + 15y - 30 = 0$$

4. ✗

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Maths - Question 46

Original paper page 37; original question and option regions

The combined equation of the bisectors of the angles between the lines joining the origin to the points of intersection of the curve $x^2 + y^2 + xy + x + 3y + 1 = 0$ and the line $x + y + 2 = 0$ is

వక్రము $x^2 + y^2 + xy + x + 3y + 1 = 0$ మరియు రేఖ $x + y + 2 = 0$ ల యొక్క ఖండన బిందువులను మూల బిందువుకు కలపగా ఏర్పడే రేఖల మధ్య గల కోణముల సమద్విఖండరేఖల ఉమ్మడి సమీకరణము

Options:

$$x^2 + 4xy - y^2 = 0$$

1. ✓

$$x^2 - 4xy + y^2 = 0$$

2. ✗

Maths - Question 46 (continued)

Original paper page 38; original question and option regions



$$2x^2 - 3xy + y^2 = 0$$

$$x^2 + 2xy - 3y^2 = 0$$

Maths - Question 47

Original paper page 38; original question and option regions

The circumference of a circle passing through the point $(4, 6)$ with two normals represented by

$$2x - 3y + 4 = 0 \text{ and } x + y - 3 = 0 \text{ is}$$

$(4, 6)$ బిందువుగుండా పోతూ, $2x - 3y + 4 = 0$ మరియు $x + y - 3 = 0$ సూచించే రేఖలను అభిలంబరేఖలుగా గల వృత్తం యొక్క చుట్టుకొలత

Options :

1. ✖ 5π

2. ✔ 10π

3. ✖ 25π

4. ✖ 8π

Mandatory : No Calculator : None Response Time : N.A Think Time : N.A 
Time : 0

If the line through the point $P(5, 3)$ meets the circle $x^2 + y^2 - 2x - 4y + \alpha = 0$ at $A(4, 2)$ and

$B(x_1, y_1)$, then $PA \cdot PB$ is equal to

$P(5, 3)$ అనే బిందువు గుండా వెళ్ళే రేఖ, $A(4, 2)$ మరియు $B(x_1, y_1)$ ల వద్ద

$x^2 + y^2 - 2x - 4y + \alpha = 0$ వృత్తాన్ని ఖండిస్తే, అప్పుడు $PA \cdot PB =$

Options :

1. ✖ 6

2. ✖ 12

3. ✖ 9

4. ✔ 8

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Maths - Question 49

Original paper page 39; original question and option regions

Consider the point $P(\alpha, \beta)$ on the line $2x + y = 1$. If the P and $(3, 2)$ are conjugate points with respect to the circle $x^2 + y^2 = 4$, then $\alpha + \beta =$

$2x + y = 1$ అనే రేఖపై ఉన్న ఒక బిందువు $P(\alpha, \beta)$ ను పరిశీలిద్దాము. P మరియు $(3, 2)$ లు $x^2 + y^2 = 4$ వృత్తం దృష్ట్యా సంయుగ్మ బిందువులైతే, అప్పుడు $\alpha + \beta =$

Maths - Question 49 (continued)

Original paper page 40; original question and option regions

3

−1

−5

7

Maths - Question 50

Original paper page 40; original question and option regions

If $(1, 3)$ is the midpoint of a chord of the circle $x^2 + y^2 - 4x - 8y + 16 = 0$, then, the area of the triangle formed by that chord with the coordinate axes is

$(1, 3)$ అనే బిందువు $x^2 + y^2 - 4x - 8y + 16 = 0$ అనే వృత్తానికి గల ఒక జ్యామధ్య బిందువు అయితే, ఆ జ్యా నిరూపక అక్షాలతో ఏర్పరిచే త్రిభుజ వైశాల్యం

Options :

1. ✖ 16

2. ✔ 8

3. ✖ 4

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Maths - Question 50 (continued)

Original paper page 41; original question and option regions

$$8\sqrt{2}$$



Maths - Question 51

Original paper page 41; original question and option regions

If the circles $x^2 + y^2 + 2ax + 2y - 8 = 0$ and $x^2 + y^2 - 2x + \alpha y - 14 = 0$ intersect orthogonally, then the distance between their centres is

$x^2 + y^2 + 2ax + 2y - 8 = 0$ మరియు $x^2 + y^2 - 2x + \alpha y - 14 = 0$ వృత్తాలు పరస్పరం లంబంగా ఖండించుకుంటే, ఆ వృత్త కేంద్రాల మధ్య దూరం

Options :

$$\sqrt{242}$$

1. ✖

$$\sqrt{970}$$

2. ✖

$$\sqrt{629}$$

3. ✔

$$\sqrt{541}$$

4. ✖

Maths - Question 52

Original paper page 42; original question and option regions



If P is a point which divides the line segment joining the focus of the parabola $y^2 = 12x$ and a point on the parabola in the ratio 1:2. Then, the locus of P is

$y^2 = 12x$ పరావలయం యొక్క నాభీ మరియు పరావలయం పై గల ఒక బిందువును కలిపే రేఖాఖండాన్ని P అనే బిందువు 1:2 నిష్పత్తిలో విభజిస్తుంటే P యొక్క బిందుపథం

Options :

1. ✖ $y^2 = 2(x - 2)$

2. ✖ $y^2 = 4x$

3. ✔ $y^2 = 4(x - 2)$

4. ✖ $y^2 = 9(x - 3)$

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Maths - Question 53

Original paper page 42; original question and option regions

Let T_1 be the tangent drawn at a point $P(\sqrt{2}, \sqrt{3})$ on the ellipse $\frac{x^2}{4} + \frac{y^2}{6} = 1$. If (α, β) is the point

where T_1 intersects another tangent T_2 to the ellipse perpendicularly, then $\alpha^2 + \beta^2 =$

T_1 అనేది $\frac{x^2}{4} + \frac{y^2}{6} = 1$ దీర్ఘవృత్తంపై $P(\sqrt{2}, \sqrt{3})$ వద్ద గీయబడిన స్పర్శరేఖ. T_1 ను

(α, β) బిందువు వద్ద ఈ దీర్ఘవృత్తానికి గల మరో స్పర్శరేఖ T_2 లంబంగా ఖండిస్తే, $\alpha^2 + \beta^2 =$

Maths - Question 53 (continued)

Original paper page 43; original question and option regions

10

52

26

$\frac{5}{12}$

Maths - Question 54

Original paper page 43; original question and option regions

If $y = x + \sqrt{2}$ is a tangent to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{2} = 1$, then equations of its directrices are

$\frac{x^2}{a^2} - \frac{y^2}{2} = 1$ అతిపరావలయం యొక్క ఒక స్పర్శరేఖ $y = x + \sqrt{2}$ అయితే దాని నియతరేఖల
యొక్క సమీకరణాలు

Options :

$$x = \pm\sqrt{3}$$

1. ✖

$$x = \pm\sqrt{\frac{8}{3}}$$

2. ✔

Maths - Question 54 (continued)

Original paper page 44; original question and option regions

$$x = \pm \sqrt{\frac{2}{3}}$$

$$x = \pm \sqrt{\frac{4}{3}}$$

4. ✖



The area of the quadrilateral formed with the foci of the hyperbola $\frac{x^2}{16} - \frac{y^2}{9} = 1$ and its conjugate hyperbola is (in square units)

$\frac{x^2}{16} - \frac{y^2}{9} = 1$ అతిపరావలయము మరియు దాని సంయుగ్మ అతిపరావలయము యొక్క నాభులతో ఏర్పడే చతుర్భుజం యొక్క వైశాల్యం (చ. యూనిట్లలో)

Options :

1. ✖ 24

2. ✖ 16

3. ✖ 25

4. ✔ 50

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Maths - Question 55 (continued)

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